

TECHNICAL REPORT

AI-Powered Smart Agriculture Monitoring System Using
Drones and Artificial Intelligence

TRINETRA DRONES AND ROBOTICS PVT.LTD

JULY 2026



ANSHIKA KODUKULA

SRM UNIVERSITY - AP24110020029

1. Objective

The objective of Week 5 was to enhance the smart agriculture monitoring system by integrating artificial intelligence modules, improving system intelligence, and developing a prototype dashboard for demonstrating crop monitoring, disease detection, pest identification, and yield prediction capabilities.

2. Technologies Used

Technology	Purpose
Python	Backend development
Streamlit	Dashboard and user interface
AI Models (Conceptual)	Disease detection and forecasting
Drone Imaging	Crop monitoring
GitHub	Version control and project management

3. System Overview

The proposed system utilizes drones to capture agricultural images and environmental data. The collected information is processed using artificial intelligence techniques to assist farmers in making informed decisions regarding crop health, pest management, and productivity.

4. AI Integration Modules

4.1 Crop Disease Detection

-
- Technique: CNN-based image classification (proposed).
 - Function: Identifies healthy and diseased crops.

4.2 Pest Detection

- Technique: YOLO-based object detection (proposed).
- Function: Detects and counts pests in crop fields.

4.3 Yield Prediction

- Technique: LSTM-based forecasting (proposed).
- Function: Predicts future crop yield using historical data.

5. System Architecture

Drone → Image Acquisition → AI Processing →

Dashboard → Farmer Recommendation System

6. Workflow

Upload Drone Image

↓

AI Processing

↓

Disease Prediction

↓

Pest Monitoring

↓

Yield Prediction

↓

Recommendations

7. Prototype Implementation

A prototype dashboard was developed using Python and Streamlit.

Features Implemented

- Drone image upload interface.
- Disease prediction simulation.
- Pest monitoring simulation.
- Yield prediction simulation.
- Farmer recommendation display.

8. Demonstration Results

The prototype successfully demonstrated:

- Image upload functionality.
- AI prediction workflow.
- Dashboard visualization.
- Recommendation generation.

(Insert screenshots here.)

9. Performance Analysis

Parameter	Result
Dashboard Response Time	< 2 seconds
Image Upload Success Rate	100%
System Stability	Stable
Prototype Status	Functional

10. Challenges Faced

- Limited availability of real agricultural datasets.
- Time constraints for training deep learning models.
- Integration of multiple AI modules within the prototype.

11. Future Enhancements

- Integration of real CNN, YOLO, and LSTM models.
- Real-time drone communication.
- Cloud-based data storage.
- Advanced predictive analytics and notifications.

12. Conclusion

Week 5 successfully demonstrated the integration of artificial intelligence concepts into the smart agriculture monitoring system. A functional prototype dashboard was developed to validate the proposed architecture and workflow. The system establishes a

foundation for future implementation of real-time AI-driven precision agriculture solutions.